

The tangential $L \log L$ estimate fails at the printed $n = 2$ endpoint

A counterexample to (4.1) in arXiv:1210.1454

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Status. Candidate counterexample, likely valid, pending human review. Estimate (4.1), printed without a restriction beyond $n \geq 2$, is false already for $n = 2$, even on a compact set strictly inside a smooth domain and for a sequence converging strongly to zero in $W^{1,1}$. The case $n \geq 3$ is not settled.

1 Source question

Kałamajska, Krömer, and Kružík ask whether the nonnegative tangential Jacobian

$$\det' \xi = \det(\xi_{ij})_{i,j=1,\dots,n-1}, \quad \xi \in \mathbb{R}^{(n-1) \times n},$$

obeys the uniform estimate

$$\sup_k \int_K \gamma(\det' \nabla u_k) dx \leq C(K, \|u_k\|_{W^{1,n-1}(\Omega; \mathbb{R}^{n-1})}), \quad \gamma(s) = s \log^+ s, \quad (1)$$

for compact $K \subset \bar{\Omega}$ away from boundary points whose outer normal is not e_n [1, Section 4, equation (4.1), p. 12]. The source calls this an open problem and does not exclude $n = 2$; its following Example 4.1 explicitly starts with $n \geq 2$.

4 Higher Integrability

By a result of S. Müller [26, 27], for any bounded sequence $(u_k \subset W^{1,n}(\Omega; \mathbb{R}^N))$ with $\det \nabla u_k \geq 0$ a.e. in Ω , $\det \nabla u_k$ is locally bounded in the class $L \log L$, i.e.,

$$\sup_k \int_K \gamma(\det \nabla u_k) dx < \infty, \quad \text{with } \gamma(s) := s \ln^+ s,$$

for every $K \subset \subset \Omega$, with $\ln^+$ denoting the positive part of the logarithm. It is natural to ask whether an analogous result holds for a null Lagrangian at the boundary in place of the determinant, up to the boundary (i.e., with Ω instead of K). The example closest to Müller's original result for the determinant is the function $\det'$, given by

$$\det' \xi := \det(\xi_{ij})_{i,j=1,\dots,n-1}, \quad \text{for } \xi \in \mathbb{R}^{(n-1) \times n}.$$

This is a null Lagrangian at the boundary, at every boundary point with the normal $\varrho = e_n$. A strict analogue for the estimates in [26, 27] in this case would be an inequality as follows:

$$\sup_k \int_K \gamma(\det' \nabla u_k) dx \leq C(K, \|u\|_{W^{1,n-1}(\Omega; \mathbb{R}^{n-1})}) \quad \text{with } \gamma(s) := s \ln^+ s, \quad (4.1)$$

with a continuous function $C(K, \cdot)$, for every compact $K \subset \bar{\Omega}$ having a positive distance to the set $\{x \in \partial\Omega; \rho(x) \neq e_n\}$ where $\rho(x)$ denotes the outer unit normal to $\partial\Omega$ at x . However, it seems that it is not possible to extend Müller's proof to this case, at least not in a straightforward way, and the validity of (4.1) remains an open problem.

On the other hand, $\det' \nabla u_k$ only depends on the derivatives of u_k with respect to first $n-1$ variables. The

For a uniform estimate, the right side is naturally read as depending on a uniform Sobolev bound. The construction also defeats the per-index reading: its Sobolev norms tend to zero while the integrals tend to infinity.

2 Proof intuition

At $n = 2$, the matrix in $\det'$ is 1×1 :

$$\det' \nabla u = \partial_1 u, \quad u : \Omega \rightarrow \mathbb{R}.$$

No multidimensional determinant compensation remains. Put a positive slope A_k on an interval of width ℓ_k . Its $W^{1,1}$ cost is $A_k \ell_k$, whereas its $L \log L$ cost is $A_k \ell_k \log A_k$. Taking $A_k = e^{k^2}$ and $A_k \ell_k = 1/k$ makes the first vanish and the second equal k .

3 Counterexample

Theorem 1. *Let $\Omega = B_2(0) \subset \mathbb{R}^2$ and*

$$K = [-1/4, 1/4] \times [-1/2, 1/2] \Subset \Omega.$$

There are $u_k \in W^{1,1}(\Omega)$, $k \geq 2$, such that

$$u_k \rightarrow 0 \text{ strongly in } W^{1,1}(\Omega), \quad \det' \nabla u_k \geq 0 \text{ a.e.,}$$

but

$$\int_K \gamma(\det' \nabla u_k) dx = k \rightarrow \infty.$$

Consequently, the unqualified estimate (1) is false.

Proof. Set

$$A_k = e^{k^2}, \quad \ell_k = \frac{1}{k A_k} = \frac{e^{-k^2}}{k},$$

and define $u_k(x_1, x_2) = \psi_k(x_1)$, where

$$\psi_k(t) = \begin{cases} 0, & t \leq 0, \\ A_k t, & 0 < t < \ell_k, \\ 1/k, & t \geq \ell_k. \end{cases}$$

Almost everywhere,

$$\partial_1 u_k = A_k \mathbf{1}_{(0, \ell_k)}(x_1), \quad \partial_2 u_k = 0.$$

Thus $\det' \nabla u_k = \partial_1 u_k \geq 0$.

Since $0 \leq u_k \leq 1/k$ and $|\Omega| = 4\pi$,

$$\|u_k\|_{L^1(\Omega)} \leq \frac{4\pi}{k}.$$

Every vertical section of $B_2(0)$ has length at most 4, whence

$$\|\nabla u_k\|_{L^1(\Omega)} \leq 4 A_k \ell_k = \frac{4}{k}.$$

Therefore $\|u_k\|_{W^{1,1}} \leq (4\pi + 4)/k \rightarrow 0$.

For $k \geq 2$, $\ell_k < 1/4$, so $(0, \ell_k) \times [-1/2, 1/2] \subset K$. Since $\log A_k = k^2$,

$$\int_K \gamma(\det' \nabla u_k) dx = \ell_k A_k \log A_k = \frac{1}{k A_k} A_k k^2 = k.$$

Finally $K \Subset \Omega$, so it has positive distance from the whole boundary and automatically meets the source condition on boundary normals. $\square$

This is stronger than a bounded-sequence counterexample: adding weak convergence to the question would not repair the endpoint.

4 Scope and upgrade attempts

The construction exploits the collapse of $\det'$ to a first derivative at $n = 2$. It neither contradicts Mueller's theorem for genuinely multidimensional nonnegative Jacobians nor disproves (1) for $n \geq 3$.

Five higher-dimensional routes were checked. The source anisotropic example has a nonintegrable normal derivative. Truncating that singularity makes the normal-derivative cost suppress the logarithmic divergence. Reflection across a flat boundary leaves a codimension-one minor, not the full Jacobian covered by Mueller's theorem. A diagonal embedding of the scalar spike forces the L^{n-1} energy to dominate its determinant mass. Separated-variable and shrinking-bubble scalings encounter the same tradeoff. No proof or counterexample resulted for $n \geq 3$.

The final upgrade strengthened the low-dimensional example: choosing $A_k \ell_k = 1/k$ rather than a fixed mass makes $u_k \rightarrow 0$ strongly while $A_k \ell_k \log A_k \rightarrow \infty$. This rules out weak-convergence and boundedness repairs of the printed statement.

5 Novelty and verification

A bounded primary-source search used the exact source title, equation (4.1), the phrases higher integrability and null Lagrangians at the boundary, and the terms tangential Jacobian and $L \log L$. Twelve indexed later citations through 2020 were screened by title and primary text where available; none explicitly resolves (4.1). No later primary statement of this $n = 2$ observation was found. Novelty confidence is moderate-low because the example is elementary; correctness confidence is high.

The included verifier checks the exact scaling formulas. Human review should first decide whether the source intended its printed $n \geq 2$ range. If it silently intended $n \geq 3$, this packet is a scope correction, not a solution of that higher-dimensional problem.

References

- [1] A. Kałamajska, S. Krömer, and M. Kružík, *Sequential weak continuity of null Lagrangians at the boundary*, Calc. Var. Partial Differential Equations 49 (2014), 1263–1278; arXiv:1210.1454.